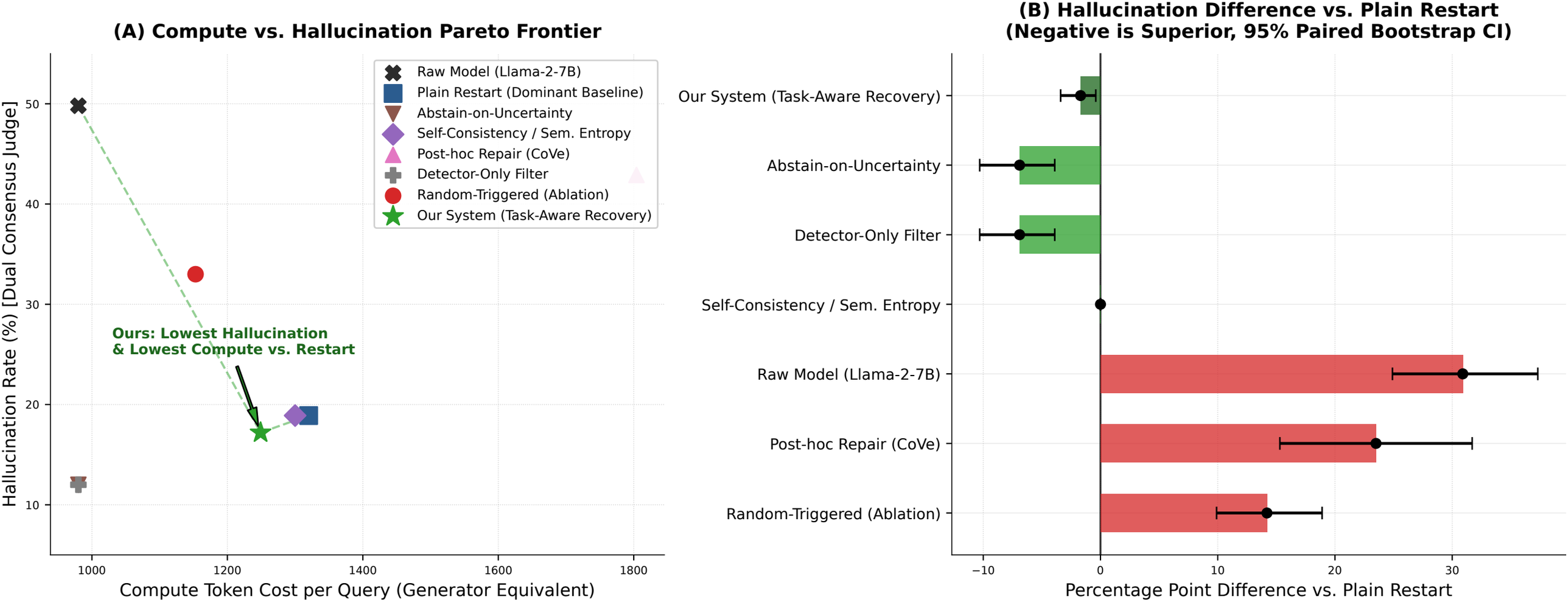
(A) Compute vs. Hallucination Pareto Frontier
Raw Model (Llama-2-7B)
Plain Restart (Dominant Baseline)
Abstain-on-Uncertainty
Self-Consistency / Sem. Entropy
Post-hoc Repair (CoVe)
Detector-Only Filter
Random-Triggered (Ablation)
Our System (Task-Aware Recovery)
Ours: Lowest Hallucination
& Lowest Compute vs. Restart
Hallucination Rate (%) [Dual Consensus Judge]
Compute Token Cost per Query (Generator Equivalent)
(B) Hallucination Difference vs. Plain Restart
(Negative is Superior, 95% Paired Bootstrap CI)
Our System (Task-Aware Recovery)
Abstain-on-Uncertainty
Detector-Only Filter
Self-Consistency / Sem. Entropy
Raw Model (Llama-2-7B)
Post-hoc Repair (CoVe)
Random-Triggered (Ablation)
Percentage Point Difference vs. Plain Restart